

- 1) ac-dc conversion systems esp. the full wave bridge. Half-wave full-wave 的公式要记住

⑥ Rectifier Comparison.

(1) Assuming the same R & C

Half-wave

Full-wave

$$V_r = (V_p - V_{on}) \frac{T}{RC} \quad > \quad V_r = (V_p - V_{on}) \frac{T}{2RC}$$

$$\Delta T = \frac{1}{\omega} \sqrt{2 \frac{V_r}{V_p}} \text{ [sec]} \quad > \quad \Delta T = \frac{1}{\omega} \sqrt{2 \frac{V_r}{V_p}} \text{ [sec]}$$

$$I_p = \frac{V_p - V_{on}}{R} \frac{2T}{\Delta T} \quad > \quad I_p = \frac{V_p - V_{on}}{R} \frac{T}{\Delta T}$$

(2) Assuming the same ripple voltage (V_r)

Half-wave

$$C = \frac{V_p - V_{on}}{V_r} \frac{T}{R}$$

$$PIV = 2V_p$$

$$I_p = \frac{V_p - V_{on}}{R} \frac{2T}{\Delta T}$$

Full-wave

$$C = \frac{V_p - V_{on}}{V_r} \frac{T}{2R}$$

$$PIV = 2V_p$$

$$I_p = \frac{V_p - V_{on}}{R} \frac{T}{\Delta T}$$

Full-wave Bridge

$$C = \frac{V_p - 2V_{on}}{V_r} \frac{T}{2R}$$

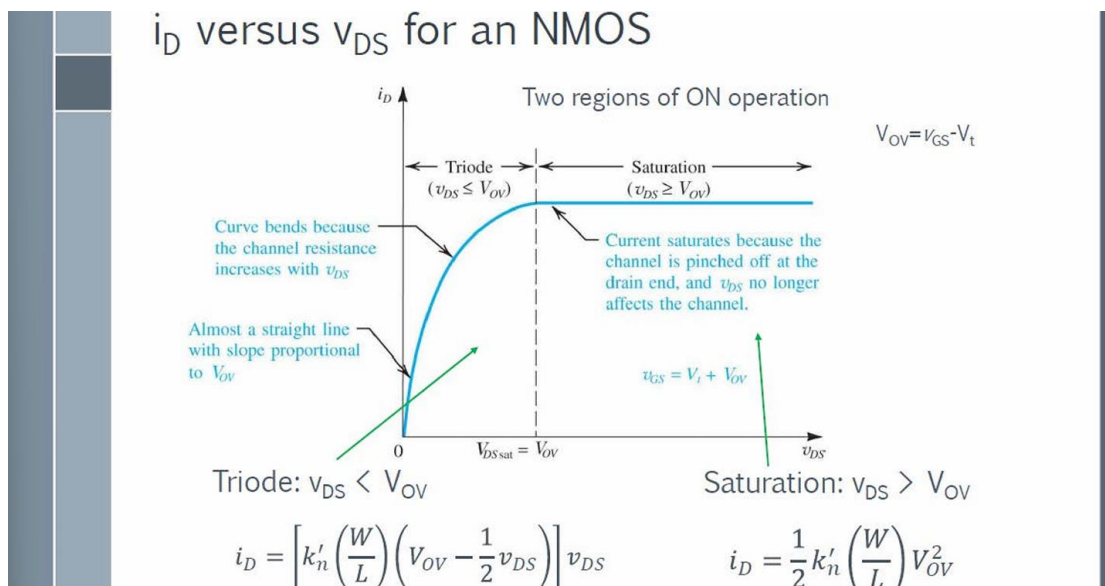
$$PIV = V_p$$

$$I_p = \frac{(V_p - 2V_{on}) * T}{(R * \Delta T)}$$

Quantity	Half-wave rectifier	Full-wave rectifier
Time between charging peaks T_r	T	$\frac{T}{2}$
Charging time / diode ON time ΔT	$\Delta T = \frac{1}{\omega} \sqrt{\frac{2V_R}{V_p}}$	$\Delta T = \frac{1}{\omega} \sqrt{\frac{2V_R}{V_p}}$

Quantity	Half-wave rectifier	Full-wave rectifier
Discharging time t_1	$t_1 = T - \Delta T$	$t_1 = \frac{T}{2} - \Delta T$
Ripple voltage V_R	$V_R \approx V_M(1 - e^{-(T-\Delta T)/(RC)})$	$V_R \approx V_M(1 - e^{-(T/2-\Delta T)/(RC)})$
Ripple voltage approximation	$V_R \approx \frac{V_M(T - \Delta T)}{RC} \approx \frac{V_M T}{RC}$	$V_R \approx \frac{V_M(T/2 - \Delta T)}{RC} \approx \frac{V_M T}{2RC}$
DC output voltage V_{DC}	$V_{DC} \approx V_M - \frac{V_R}{2}$	$V_{DC} \approx V_M - \frac{V_R}{2}$
DC load current I_{DC}	$I_{DC} \approx \frac{V_{DC}}{R}$	$I_{DC} \approx \frac{V_{DC}}{R}$
Conduction angle θ_c under your notation	$\theta_c = \omega \Delta T \approx \sqrt{\frac{2V_R}{V_p}}$	$\theta_c = \omega \Delta T \approx \sqrt{\frac{2V_R}{V_p}}$

2) MOSFET and in the circuit analysis. 线性区域的 rd



线性区，NMOS 电流公式是：

$$I_D = k_n \left[(V_{GS} - V_t) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

令：

$$V_{OV} = V_{GS} - V_t$$

则：

$$I_D = k_n \left[V_{OV} V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

小信号 drain resistance 定义为：

$$r_d = \left(\frac{\partial I_D}{\partial V_{DS}} \right)^{-1}$$

对 V_{DS} 求导:

$$\frac{\partial I_D}{\partial V_{DS}} = k_n(V_{OV} - V_{DS})$$

所以线性区的小信号电阻是:

$$r_d = \frac{1}{k_n(V_{OV} - V_{DS})}$$

所以也可以写成:

$$r_d = \frac{1}{\mu_n C_{ox} \frac{W}{L} [(V_{GS} - V_t) - V_{DS}]}$$

3) BJT and in the circuit analysis. 要检查工作区域

4) BJT amplifiers. 记住 step

5) CMOS digital applications. PUN 是用的反向

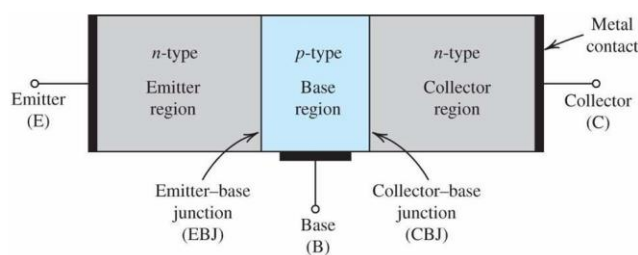
Lecture 1

Device structure and physics

> BJT consists of three semiconductor regions:

- **Emitter** region: n-type
- **Base** region: p-type
- **Collector** region: n-type

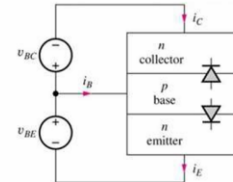
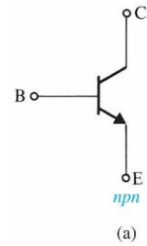
> Type described is npn. pnp types exist as well.



A simplified structure of the npn transistor

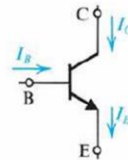
Modes of operation

- › Two pn junctions:
 - Base to emitter
 - Base to collector
- › Three modes of operation:
 - **Active** mode – used primarily for **amplification**
 - **Saturation** mode – used for primary for **switching**
 - **Cutoff** mode – device is off
- › Operating mode depends on biasing
- › Two external voltage sources are required for biasing
 - A source at the base (to bias B-E junction)
 - A source at the collector (to bias B-C junction)



Collector current in the forward active region

- › The current across a forward-biased pn junction is proportional to $e^{\frac{v_{BE}}{V_T}}$
 - where v_{BE} is the forward voltage across the base-emitter junction
 - and V_T = thermal voltage = $kT/q = 0.025$ V at room temperature



$$i_C = I_S e^{\frac{v_{BE}}{V_T}}$$

- where I_S = saturation current; typically 10^{-18} A < I_S < 10^{-9} A. I_S is proportional to the area of the B-E junction and inversely proportional to the width, W , of the base region.

Note that i_C is independent of v_{BC} in the forward active region

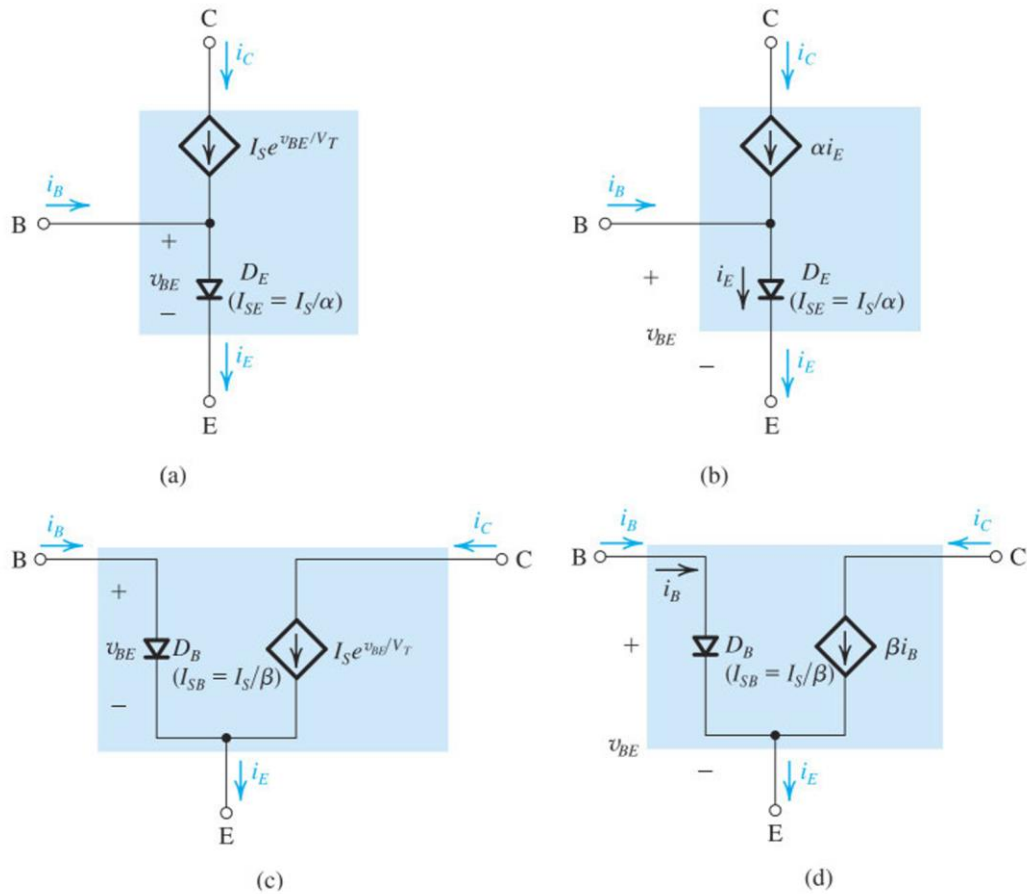


Figure 6.5 Large-signal equivalent-circuit models of the npn BJT operating in the active mode.

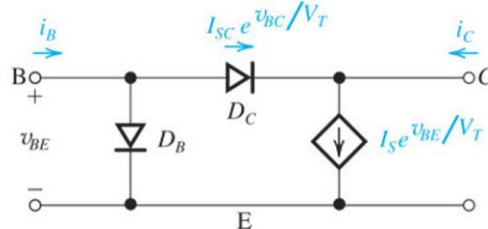
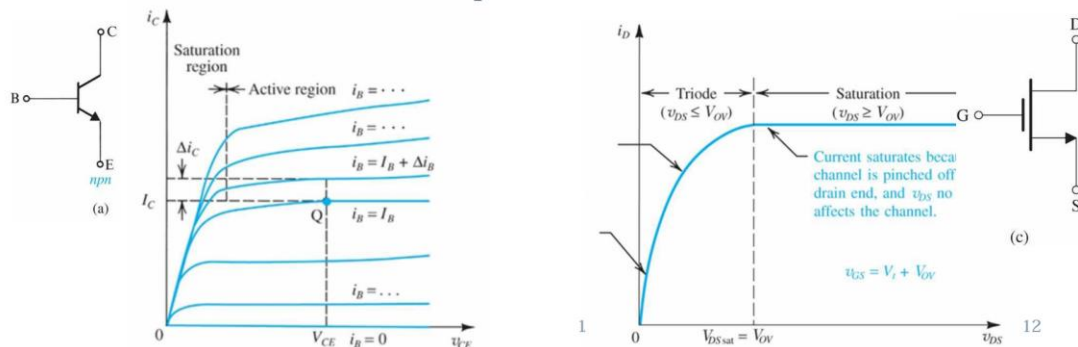


Figure 6.9 Modeling the operation of an npn transistor in saturation by augmenting the model of Fig. 6.5(c) with a forward-conducting diode D_C . Note that the current through D_C increases i_B and reduces i_C .

IV characteristics of npn BJT vs. NMOS

- Recall an NMOS with **gate**, **drain** and **source** and three operating regions: **saturation**, **triode** and **cutoff**
- Saturation mode of a BJT is completely different than saturation mode of a MOSFET
- Saturation mode of BJT corresponds to triode mode of MOSFET
- Active mode of BJT corresponds to saturation mode of MOSFET



Operational modes of the npn BJT

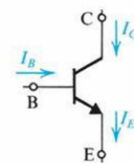
- > For BJT to operate in active mode, B-C junction must be reverse-biased
- > A pn junction does not become forward-biased until the forward voltage across it exceeds a set value (~ 0.4 V)
- > Active mode of operation of npn-BJT can be maintained down to $V_{BC} = \sim 0.4$ V, or $V_{CB} = -0.4$ V

Operation in the saturation region

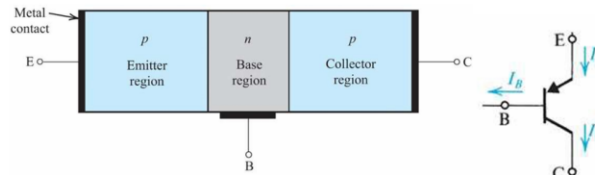
- > For a 'saturated' transistor the ratio i_C/i_B is lower than β_F
- > This ratio can be referred to 'forced' β

$$\beta_{forced} = \frac{i_C}{i_B} < \beta_F$$

- > For BJT to be in saturation:
 - B-C junction must be forward biased by more than 0.4 V

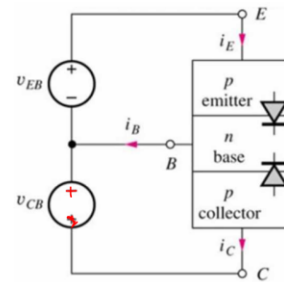


pnp transistor



- > p-type and n-type semiconductor material
- > Collector current and base current exit transistor terminals and emitter current enters the device
- > E-B junction is forward-biased and C-B junction is reverse-biased

Emitter-Base Junction	Collector-Base Junction	
	Reverse Bias	Forward Bias
Forward Bias	Forward-Active Region (Good Amplifier)	Saturation Region (Closed Switch)
Reverse Bias	Cutoff Region (Open Switch)	Reverse-Active Region (Poor Amplifier)



Summary of BJT characteristics

Table 6.2 Summary of the BJT Current-Voltage Relationships in the Active Mode

$$i_C = I_S e^{v_{BE}/V_T}$$

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta} \right) e^{v_{BE}/V_T}$$

$$i_E = \frac{i_C}{\alpha} = \left(\frac{I_S}{\alpha} \right) e^{v_{BE}/V_T}$$

Note: For the pnp transistor, replace v_{BE} with v_{EB} .

$$i_C = \alpha i_E \quad i_B = (1 - \alpha) i_E = \frac{i_E}{\beta + 1}$$

$$i_C = \beta i_B \quad i_E = (\beta + 1) i_B$$

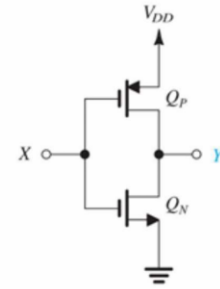
$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\alpha = \frac{\beta}{\beta + 1}$$

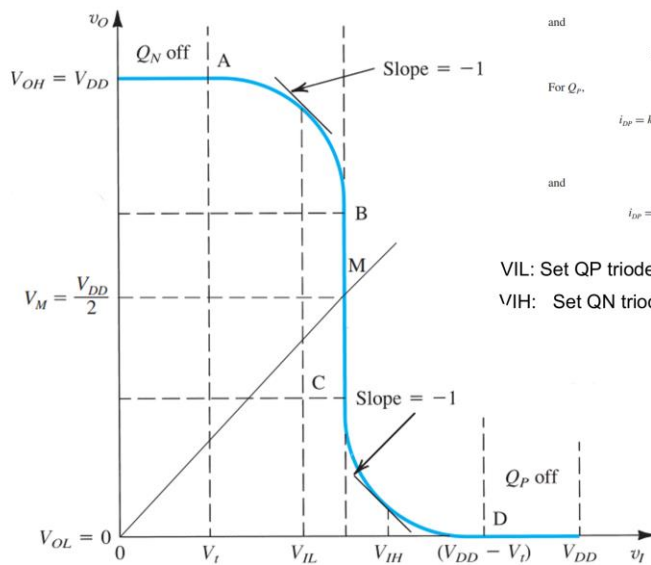
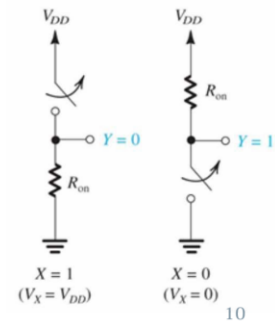
$$V_T = \text{thermal voltage} = \frac{kT}{q} \simeq 25 \text{ mV at room temperature}$$

CMOS implementation of an inverter

- CMOS inverter is shown
- It consists of an NMOS Q_N and a PMOS Q_P
- When $X=1$ or $V_X=V_{DD}$, Q_N is on and Q_P is off, $Y=0$ (V_Y is very low)
- When $X=0$ or $V_X=0$, Q_N is off and Q_P is on, $Y=1$ ($V_Y=V_{DD}$)



CMOS inverter



voltage-transfer curve. For this we need the $i-v$ relationships of Q_N and Q_P . For Q_N ,

$$i_{DN} = k_n' \left(\frac{W}{L} \right)_n \left[(v_i - V_{tn})v_o - \frac{1}{2}v_o^2 \right] \quad \text{for } v_o \leq v_i - V_{tn} \quad (14.28)$$

and

$$i_{DN} = \frac{1}{2}k_n' \left(\frac{W}{L} \right)_n (v_i - V_{tn})^2 \quad \text{for } v_o \geq v_i - V_{tn} \quad (14.29)$$

For Q_P ,

$$i_{DP} = k_p' \left(\frac{W}{L} \right)_p \left[(V_{DD} - v_i)(V_{DD} - v_o) - \frac{1}{2}(V_{DD} - v_o)^2 \right] \quad \text{for } v_o \geq v_i + |V_{tp}| \quad (14.30)$$

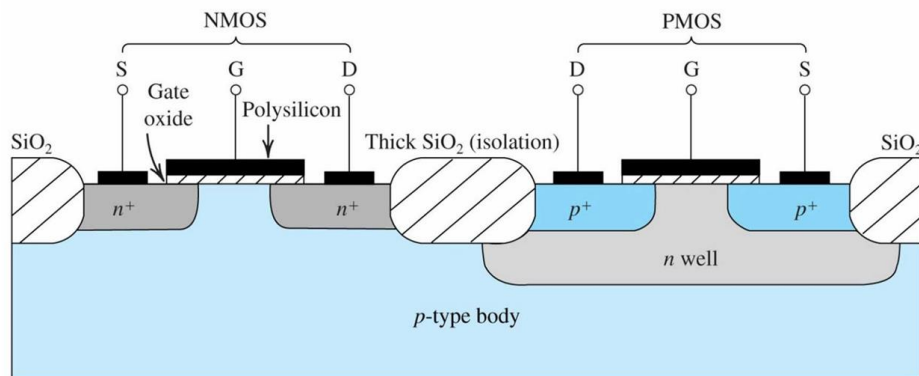
and

$$i_{DP} = \frac{1}{2}k_p' \left(\frac{W}{L} \right)_p (V_{DD} - v_i - |V_{tp}|)^2 \quad \text{for } v_o \leq v_i + |V_{tp}| \quad (14.31)$$

V_{IL}: Set QP triode current = QN saturation current

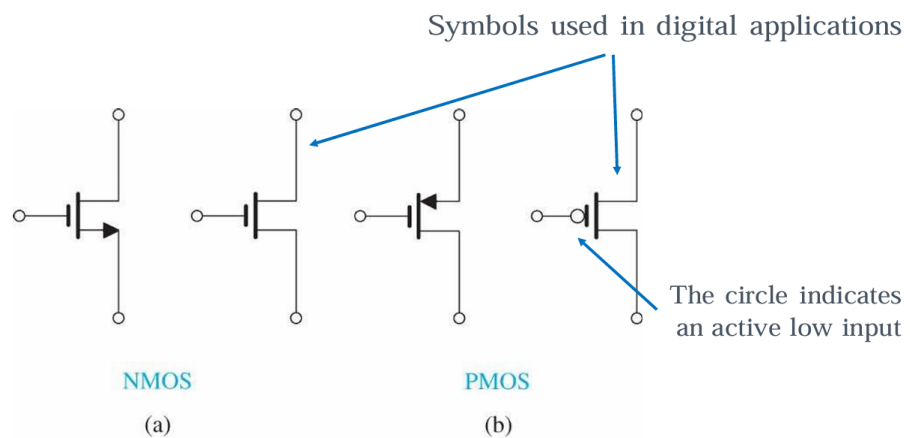
V_{IH}: Set QN triode current = QP saturation current

Complimentary MOS or CMOS



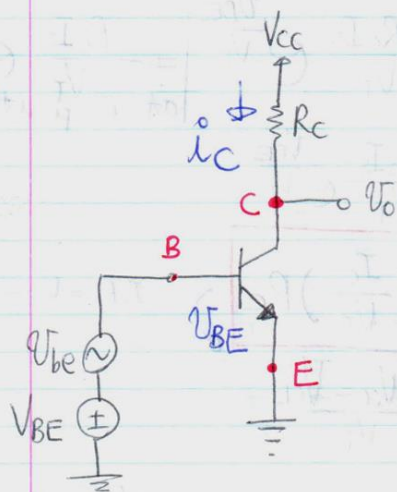
- › The NMOS transistor is implemented on a p-type substrate
- › The PMOS transistor is fabricated in a specially created n region, known as an **n well**
- › The two devices are separated from each other by an oxide layer that acts as an insulator

Alternate MOSFET symbols



Usual and alternative circuit symbols for MOSFETs.

2. Small-signal voltage gain.



$$V_{BE} = \underbrace{V_{BE}}_{\text{"DC"}} + \underbrace{v_{be}}_{\text{"AC"}}$$

= input signal

$$V_o = V_{CC} - R_C(i_c)$$

= output signal

small-signal voltage gain, A_v ,

$$A_v \equiv \left. \frac{dV_o}{dV_{BE}} \right|_{V_{BE}=V_{BE}} = \left. \frac{dV_{CE}}{dV_{BE}} \right|_{\text{at Q-point}}$$

where,

$$\begin{cases} V_{CE} = V_{CC} - R_C(i_c) \\ i_c = I_S e^{\frac{V_{BE}}{V_T}} = \underbrace{I_C}_{\text{"DC"}} + \underbrace{i_c}_{\text{"AC"}} \\ I_C = I_S e^{\frac{V_{BE}}{V_T}} \end{cases}$$

$$V_{CE} = V_{CC} - R_C I_S e^{\frac{V_{BE}}{V_T}}$$

$$\left. \frac{dV_{CE}}{dV_{BE}} = -\frac{R_C I_S}{V_T} e^{\frac{V_{BE}}{V_T}} \right|_{\text{at Q-point}}$$

$$= -\frac{R_C I_S}{V_T} e^{\frac{V_{BE}}{V_T}}$$

$$= -\left(\frac{I_C}{V_T}\right) R_C \quad \left. \begin{array}{l} I_C R_C = V_{CC} - V_{CE} \\ \end{array} \right\}$$

$$= -\frac{V_{CC} - V_{CE}}{V_T}$$

V_T = thermal voltage $\rightarrow \sim 25\text{mV}$ @ room temp.

② Collector current & the transconductance

$$\begin{aligned} i_c &= I_s e^{\frac{V_{BE}}{V_T}} \\ &= I_s e^{\frac{V_{BE} + V_{be}}{V_T}} = I_s e^{\frac{V_{BE}}{V_T}} e^{\frac{V_{be}}{V_T}} \\ &= I_c e^{\frac{V_{be}}{V_T}} \quad \left. \vphantom{I_c} \right\} \text{if } V_{be} \ll V_T, \\ &\approx I_c \left(1 + \frac{V_{be}}{V_T} \right) \end{aligned}$$

Here we have expanded the exponential in a series and retained only the first two terms. That is, we have assumed that

$$V_{be} \ll V_T$$

"small-signal approximation condition for a BJT"

$\Rightarrow$ this approximation for the BJT is valid only for V_{be} less than 5 mV $\sim$ 10 mV, at most.

Under this approximation,

$$i_c = I_c + \underbrace{\left(\frac{I_c}{V_T} V_{be} \right)}_{\equiv \underline{i_c}}$$

$\Downarrow$
dc bias component

$\Downarrow$
small-signal component

$$\underline{i_c} = \frac{I_c}{V_T} \underline{V_{be}} = g_m V_{be}$$

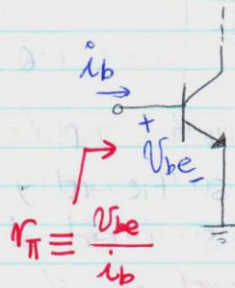
$$g_m \text{ (transconductance)} \equiv \frac{i_c}{V_{be}} = \frac{I_c}{V_T}$$

③ i_B and the input resistance at the Base.

$$\begin{aligned} i_B &= \frac{i_c}{\beta} = \frac{1}{\beta} (I_C + \underline{i_c}) \\ &= \frac{I_C}{\beta} + \frac{1}{\beta} \frac{I_C}{V_T} V_{be} \\ &= I_B \quad \underbrace{\quad}_{\underline{i_b} \text{ (signal component)}} \end{aligned}$$

$$\underline{i_b} = \frac{1}{\beta} \frac{I_C}{V_T} \underline{V_{be}} = \frac{g_m}{\beta} \underline{V_{be}}$$

$$\Rightarrow r_{\pi} \equiv \frac{V_{be}}{i_b} = \frac{\beta}{g_m} = \frac{V_T}{I_B}$$



④ i_E and the input resistance at the Emitter

$$i_E = \frac{i_c}{\alpha} = \frac{I_C}{\alpha} + \frac{\underline{i_c}}{\alpha} = I_E + \underline{i_e}$$

$$\underline{i_e} = \frac{i_c}{\alpha} = \frac{I_c}{\alpha V_T} V_{be} = \frac{I_E}{V_T} \underline{V_{be}}$$

$$\boxed{r_e \text{ (emitter resistance)}} \equiv \frac{\underline{V_{be}}}{\underline{i_e}} = \frac{V_T}{I_E}$$

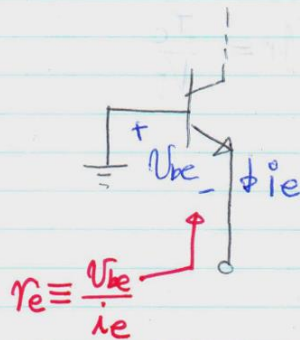
$$= \frac{\alpha V_T}{I_c} = \boxed{\frac{\alpha}{g_m}} \simeq \frac{1}{g_m}$$

$$V_{be} = i_b r_{\pi} = i_e r_e$$

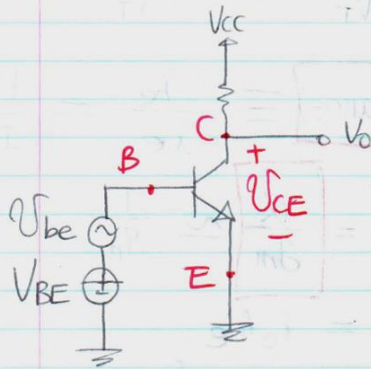
$$\text{thus, } r_{\pi} = \frac{i_e}{i_b} r_e = \frac{\frac{I_c}{\alpha V_T} V_{be}}{\frac{g_m}{\beta} V_{be}} r_e$$

$$r_{\pi} = \frac{\beta}{\alpha} r_e = \frac{\beta}{\frac{\beta}{1+\beta}} r_e = (1+\beta) r_e$$

$$\Rightarrow \boxed{r_{\pi} = (1+\beta) r_e}$$



⑤ Voltage Gain



$$\begin{aligned}V_{CE} &= V_{CC} - i_C R_C \\&= V_{CC} - (I_C + \underline{i_c}) R_C \\&= (V_{CC} - I_C R_C) - \underline{i_c R_C} \\&= V_{CE} - \underline{i_c R_C} \\&= V_{CE} + \underline{v_{ce}}\end{aligned}$$

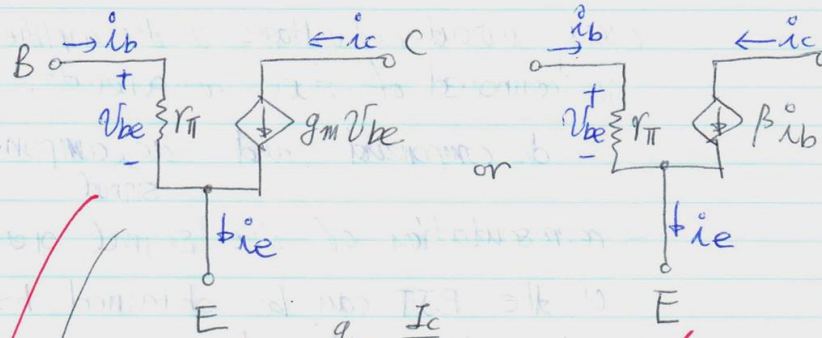
thus,

$$\begin{aligned}\underline{v_{ce}} &= -\underline{i_c R_C} = -g_m \underline{v_{be}} R_C \\&= -(g_m R_C) \underline{v_{be}}\end{aligned}$$

$$A_v \equiv \frac{v_{ce}}{v_{be}} = -g_m R_C$$

$$g_m = \frac{I_C}{V_T}$$

① The hybrid- π equivalent circuit model.



$$g_m = \frac{I_C}{V_T}$$

$$r_{\pi} = V_T / I_B = \beta / g_m$$

Can you verify that

$$\frac{v_{be}}{r_{\pi}} + g_m v_{be} = i_e \quad ?$$

This model represents the BJT as a voltage controlled current source.

current controlled current source

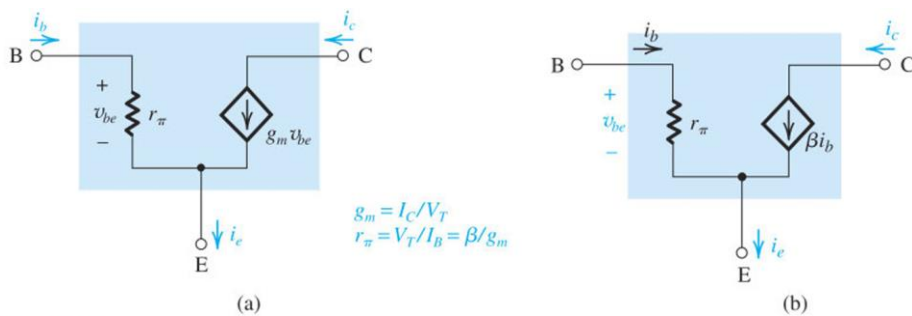
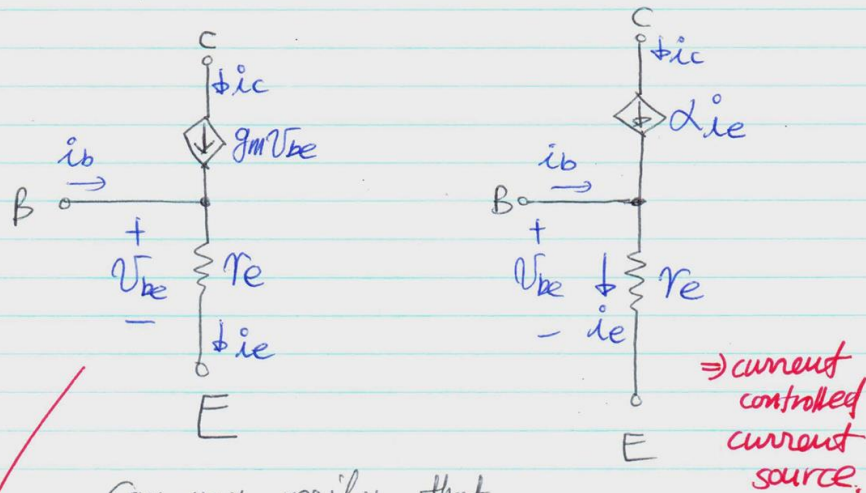


Figure 7.25 Two slightly different versions of the hybrid- π model for the small-signal operation of the BJT. The equivalent circuit in (a) represents the BJT as a voltage-controlled current source (a transconductance amplifier), and that in (b) represents the BJT as a current-controlled current source (a current amplifier).

⑧ The T-model.

Although the hybrid- π model can be used to carry out small-signal analysis of any transistor circuit, there are situations in which an alternative model, known as a T-model, is much more convenient.



Can you verify that

$$i_b + g_m v_{be} = i_e = \frac{v_{be}}{r_e} \quad ?$$

This model represents the BJT as a voltage controlled current source

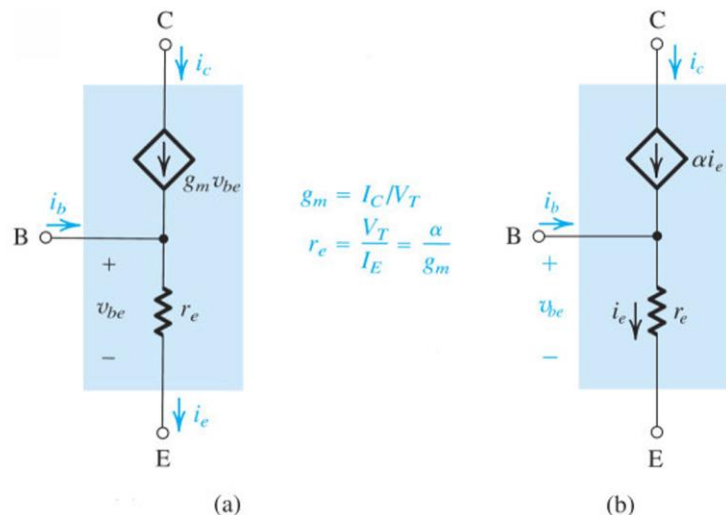


Figure 7.27 Two slightly different versions of what is known as the T model of the BJT. The circuit in (a) is a voltage-controlled current source representation and that in (b) is a current-controlled current source representation. These models explicitly show the emitter resistance r_e rather than the base resistance r_π featured in the hybrid- π model.